

# 1. Overfitting Analysis

## 1.1 The Baseline Pitfall (The Unconstrained Decision Tree)

In the initial baseline implementation using an unconstrained `DecisionTreeRegressor`, a severe generalization gap was identified. When a decision tree is allowed to grow without constraints (i.e., default parameters in scikit-learn), it splits nodes until all leaves are pure or contain fewer than the minimum sample threshold.

- **Training Performance:** The training Root Mean Squared Error (RMSE) approached a near-zero or extremely low value ( $RMSE_{\text{train}} \approx 0.0$ ), while the  $R^2$  score neared 1.0. This indicates that the model perfectly memorized the noise and spatial anomalies inherent in the California Housing Dataset.
- **Test Performance:** When evaluated on the unseen test partition (20% validation split), the RMSE spiked significantly higher, and the  $R^2$  dropped drastically relative to the training phase.

**Interpretation:** This substantial divergence between training error and testing error is the classic signature of **high variance / overfitting**. The model built overly complex, fragmented decision boundaries that fail to map effectively onto unseen data distributions.

## 1.2 Cross-Validation Stability

To accurately gauge the model's actual predictive capability beyond a single lucky or unlucky train-test partition, a 5-Fold Cross-Validation strategy was applied using the negative root mean squared error metric:

$$CV \text{ RMSE} = \frac{1}{K} \sum_{i=1}^K RMSE_i$$

The cross-validation score  $CV_{\text{rmse}}$  provided a stable, realistic estimation of performance. It proved that the baseline decision tree's stellar performance was an illusion created by training data memorization, confirming the critical industry practice of relying on multi-split validation prior to production deployment.

# 2. Hyperparameter Tuning Approach

To mitigate high variance and control the model's complexity, systematic hyperparameter optimization was conducted using `GridSearchCV` combined with 5-Fold Cross-Validation.

## 2.1 Hyperparameter Space & Intuition

The optimization targeted the foundational structural constraints of the tree architecture:

- `max_depth` [3, 5, 7, 10]: Limits how deep the tree can grow. Shallow depths prevent the tree from capturing complex interactions (underfitting), whereas excessive depths cause the model to capture sample-specific noise (overfitting).
- `min_samples_split` [2, 5, 10]: Specifies the minimum number of samples required to split an internal node. Higher values restrict the formation of highly specific leaves, smoothing out the prediction surfaces.

## 2.2 Execution Workflow

The search grid evaluated all  $4 \times 3 = 12$  unique parameter combinations across 5 distinct data folds, resulting in a total of 60 individual model fits. The objective function maximized the negative RMSE, directly translating to minimizing the true generalization error.

Raw Data → Feature Scaling → 5-Fold CV Split → Parameter Grid Evaluation → Optimal Selection

## 3. Model Comparison & Final Selection Justification

### 3.1 Performance Matrix

The table below outlines the performance metrics collected across the different iterations, comparing the baseline models from Task-2 against the newly optimized candidate.

| Model Evaluation Criteria          | Validation RMSE   | R2 Score            | Overfitting Tendency               |
|------------------------------------|-------------------|---------------------|------------------------------------|
| Linear Regression (Baseline)       | Moderate / High   | Baseline            | Low Variance / High Bias           |
| Ridge Regression (Task-2)          | Moderate          | Marginally Improved | Well-regulated                     |
| Unconstrained Decision Tree        | High (Test Split) | Poor (Test Split)   | Severe Overfitting / High Variance |
| Tuned Decision Tree (GridSearchCV) | Lowest (Balanced) | Highest (Balanced)  | Controlled / Stable                |

### 3.2 Final Selection Justification

The **Tuned Decision Tree** is selected as the optimal production-ready model for the California Housing Dataset based on the following justifications:

1. **Reduction of Overfitting:** By enforcing optimal structural limits (e.g., constraining `max_depth` and raising `min_samples_split`), the massive performance gap between the training set and the validation set was closed. The model shifted along the **Bias-Variance Tradeoff** curve away from high variance towards a highly stable equilibrium.
2. **Superior Generalization:** The Tuned Decision Tree outperforms both Linear and Ridge Regressions by safely capturing non-linear relationships and interactions among features (such as geographical coordinates and income-to-population dynamics) without succumbing to memorization.
3. **Cross-Validation Reliability:** Because the performance scores are validated across 5 distinct data folds, we possess high empirical confidence that the model will maintain its predictive accuracy on live, unseen operational data without suffering production degradation.